Quoc Chuong Nguyen

Curriculum Vitae

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Google Scholar	: https://scholar.google.com/citations?user=embX1-gAAAAJ&hl=en
GitHub	: https://github.com/ChuongQuoc1413017

Research Interest

Working on quantum algorithms and quantum machine learning, focusing on benchmarking, verification, and scalable simulation, with applications to quantum many-body systems and complexity analysis

Mathematics	: Game Theory · Network Science · Category Theory
Computer Science	: Complexity Theory · Property Testing · Machine Learning
Quantum Computation	: Quantum Algorithm · Quantum Artificial Intelligence

Publication

Published

- **(First author) Chuong** et al. "Discovery of connectivity-trainability trade-off of IQP circuits for Hamiltonian optimization", Physica Scripta, 2026
- **(Co-author) Chuong** et al. "FQsun: A Configurable Wave Function-Based Quantum Emulator for Power-Efficient Quantum Simulations", IEEE Access, vol. 13, pp. 93271-93286, 2025
- **(First-author) Chuong** et al. "Qsun: an open-source platform towards practical quantum machine learning applications", Machine Learning: Science and Technology, vol. 3, pp. 015034, 2022

Preprints

- (First-author) Chuong. "Network Sampling: An Overview and Comparative Analysis", arXiv:2504.17701, 2025
- **(Co-author) Chuong** et al. "Hybridising Reinforcement Learning and Heuristics for Hierarchical Directed Arc Routing Problems", arXiv:2501.00852, 2025
- (First-author) Chuong. "Monte Carlo Analysis of Boid Simulations with Obstacles: A Physics-Based Perspective", arXiv:2412.10420, 2024
- **(First-author) Chuong** and Trung Kien Le. "Toward a comprehensive simulation framework for hypergraphs: a Python-base approach", arXiv:2401.03917, 2024

Academic Experience

Recent Remote Collaborations

- May 2024 - Now: In collaboration with Prof. Hy Truong Son (University of Alabama at Birmingham, US) and Prof. Thu Huong Dang (Lancaster University, England) on **Operations Research**

Reviewer for Peer-Reviewed Journals

- SoftwareX, Quantum Machine Intelligence, Scientific Reports

Research Positions

Institute of Fundamental & Applied Sciences, Duy Tan University — July 2025 - Now
Website: https://ifas.duytan.edu.vn/
Position: Researcher
Job Description: develop quantum encoding frameworks for quantum many-body systems, focusing on benchmarking and simulation efficiency on GPU and quantum hardware

QuantumLab - Ho Chi Minh City Institute of Physics, Vietnam — Sep 2021 - May 2022
Website: https://lantrann.github.io/QuantumLab-HCMIP
Position: Affiliated Member
Job Description: develop quantum machine learning applications in chemistry and quantum metrology

Research Assistant for Dr. Hung Q. Nguyen, Vietnam — Nov 2019 - Apr 2023
Position: Research Assistant
Job Description: develop Qsun (Quantum Computing simulator) and Variational Quantum Eigensolver's applications in physics

Education

Aug 2023 - Jun 2025 — **Master of Arts in Mathematics at SUNY, University at Buffalo (State University of New York at Buffalo)**, New York City, the US
Track: Applied Mathematics
CGPA: 3.889/4.0 and **Degree GPA**: 3.967/4.0 (A/A-: High Distinction)
Conferral Date: June 1, 2025

Dec 2014 - Nov 2018 — **Bachelor of Science in Physics at University of Science (HCMUS) Vietnam National University, Ho Chi Minh City (VNU-HCM)**, VN
Track: Theoretical Physics
CGPA: 8.27/10 or 3.5/4.0 (A, top 2% over 300 students)
Graduated Seminar: Simulation of Quantum Many-body Dynamics using the Monte-Carlo algorithm with Python (Non-Thesis option)
Conferral Date: November 4, 2018

Projects and Resources

Software

- QSun: Quantum platform to simulate quantum circuits using NumPy array
  - https://github.com/ChuongQuoc1413017/Quantum_Virtual_Machine
- Quantum Evolutionary Algorithm: Optimize quantum circuits' structure using Evolutionary Algorithms
  - https://github.com/ChuongQuoc1413017/Quantum_Evolutionary_Algorithm
- HyperRD: Simulation and analysis of random and dynamical hypergraph
  - https://github.com/ChuongQuoc1413017/Hypergraph_RD

Lecture Notes

- Summary of Lie Algebra in Particle Physics (Vietnamese)
  - https://github.com/ChuongQuoc1413017/Note/blob/main/Lie%20Algebra.pdf
- Quantum Machine Learning at ⟨QCS | 2021 ⟩
  - https://github.com/ChuongQuoc1413017/Note/blob/main/QML_Lecture_Note.pdf

Talks/Posters at Conferences, Workshops, and Schools

"Tutorial on Quantum Computing using Qiskit and Python" — 22 Apr - 25 Apr 2026
at School on Quantum Optimization Algorithms 2026, UPT, Vietnam — Lecturer

"Introduction to Quantum Algorithms" (Undergraduate Level) — 22 Nov 2025
at Quantum Winter School, HCMUT, Vietnam — Lecturer

"Quantum Machine Learning & Qsun platform" — 13 Oct - 15 Oct 2025
at PIPC 2025: Celebrating 100 years of quantum physics, Vietnam — Poster

"Quantum Machine Learning" — 19 Aug 2025
at Quantum Computing Seminar, Vietnam — Speaker

"Quantum Machine Learning" (Graduate Level) — 04 Dec - 12 Dec 2021
at Quantum Computing School - ⟨QCS | 2021 ⟩, Vietnam — Contributory Lecturer

"Half-wormhole solutions & Black Hole singularity" — 13 Oct 2021
at The 46th Vietnam Conference on Theoretical Physics (VCTP-46) — Poster

Honors & Awards

Fellowships & Teaching Assistantship (SUNY, University at Buffalo) — Aug 2023 - June 2025
Description: for Master/Ph.D. student with a good profile (GPA at least 3.0/4.0, at most 10 students selected) each semester — 2 academic years

Academic incentive fellowships (University of Science, HCMUS) — Nov 2014 - Nov 2018
Description: for students with the highest GPA (top 10 students) and contributions to society through activities each semester — 4 academic years

Working Experience

Teaching

Teaching Assistant (TA) at HCMUS, Vietnam (Graduate Level) — Sep 2025 - Sep 2026
Description: Instructor in Quantum Algoirthms class

Mentor of Directed Reading Project program at SUNY, UB — Jan 2024 - May 2024
Description: teach undergrad students about Computational Thinking and Data Structures & Algorithms

Teaching Assistant (TA) at SUNY, UB (Undergraduate Level) — Aug 2023 - May 2025
Description: TA in Calculus, Linear Algebra, and Stochastic Processes

Company

FUNiX Co. Ltd., Vietnam — Jul 2020 - Nov 2021
Position: Collaborative Assistant
Job Description: develop machine learning & data science courses, translate course content from Vietnamese to English, and fix errors

Saigon Scientists Co., Vietnam — Jul 2018 - Jul 2019
Position: Research & Development Assistant
Job Description: develop STEM lessons, translate courses' contents from Vietnamese to English, and project management

Languages & Computer Skills

Vietnamese:	Mother tongue
English:	Fluent (professional working proficiency)
Programming languages:	Proficient: Python, NetworkX, Qiskit, Pennylane
	Basic: Mathematica, Matlab, R, LaTeX, Haskell, Lean

Selected Academic Events Attended

CQT Quantum Hackamonth	1 - 26 Jun 2026
VIASM-ICTP Summer School in Group Theory and Representation Theory	9 - 21 Aug 2022
Topological Quantum Electrons Interacting In-persons	10 - 16 July 2022
Vietnam - USA Joint Mathematical Meeting - VNUS 2019	10 - 13 Jun 2019

Selected Certifications

- Introduction to Discrete Mathematics for Computer Science Specialization on *Mathematical Thinking, Probability, Combinatorics, and Graph Theory* at Coursera in 2022, Credential ID: ZDLRFFSJ6H8N
- Introduction to Logic and Critical Thinking Specialization on *Argument Analysis, Deductive Reasoning, Inductive Reasoning, and Fallacies* at Coursera in 2022, Credential ID: PNT6RQ3G9Q89
- Mind and Machine Specialization on *Cognitive Science and Artificial Intelligence* at Coursera in 2022, Credential ID: HNTZFNDS4LU7

References

Prof. Thu Huong Dang (Google Scholar)
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Email: hungngq@hus.edu.vn

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